

Trần Hoàng Khải

Software Engineer

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CAREER OBJECTIVE

- Software Engineer with hands-on experience building and optimizing enterprise systems using C#, ASP.NET, and SQL Server. Delivered features and fixes on a multi-tenant ERP platform, with strengths in debugging, database optimization, and system integrations. Seeking to grow further into scalable system design and distributed architectures.

EDUCATION

VNUHCM - UNIVERSITY OF SCIENCE (HCMUS) OCT. 2021 - 2025

- BACHELOR OF INFORMATION TECHNOLOGY
- Major: Information Systems

WORK EXPERIENCE

Company: EMERGE app 12/2025 - 6/2026

- **Role:** Fullstack developer
- **Product:** Emerge ERP System (Multi-tenant)
- **Working model:** Agile Scrum

Responsibilities

- Delivered 86 tickets (90.7% on-time) in 6 sprints using AngularJS, ASP.NET Web API (C#), SQL Server
- Investigated and resolved issues via end-to-end flow tracing (UI → API → database)
- Implemented PDF generation workflows (Sales Orders, Quotations, Delivery Orders) using iTextSharp with multi-language support
- Built backend features for document processing, serial number tracking, and data workflows
- Optimized frontend rendering for large product datasets via lazy loading, cutting page load time from ~30s to 2-5s
- Integrated external systems including Shopify (REST & GraphQL APIs) with background processing (Quartz.NET, MassTransit, RabbitMQ)

Achievements & Learnings

- Eliminated N+1 query issues, consolidating multiple queries into a single async LINQ projection to improve performance and reduced DB load
- Resolved multi-tab race conditions, ensuring data consistency in critical Quotation-to-SalesOrder workflows
- Enhanced frontend responsiveness using progressive rendering techniques, removing UI freezes on large datasets
- Strengthened expertise in debugging, root cause analysis, and distributed system concepts (message queues, background jobs, integrations)
- Leveraged AI-assisted tooling (Claude Code) to accelerate debugging and root cause analysis across complex workflows, as well as suggesting and evaluating solution approaches based on system requirements, improving issue resolution efficiency.

Technologies:

- **Backend:** ASP.NET Web API (C#), Entity Framework
- **Frontend:** AngularJS
- **Database:** Microsoft SQL Server
- **Messaging & Jobs:** RabbitMQ, MassTransit, Quartz.NET
- **Version Control:** Azure DevOps

PROJECT EXPERIENCE

Project: Dealership Service Scheduler 11/2026

Individual project (technical challenge)

Description:

Backend service for automotive dealership aftersales that books service appointments and automatically assigns a suitable service bay and qualified technician for the full service duration.

- Repo: github.com/THKhai/dealership-service-scheduler
- Video walkthrough: youtu.be/2LBcBDD7Feo

Responsibilities:

- Designed the system end-to-end: business process, use cases, three-layer architecture, domain model and sequence diagrams (System Design Document in repo)
- Built a REST API with .NET 10, EF Core and PostgreSQL
- Prevented double-booking at the database level with PostgreSQL exclusion constraints (tstzrange), verified by concurrent-request integration tests (xUnit + Testcontainers)
- Added structured logging (Serilog), metrics and tracing (OpenTelemetry)
- Used Claude Code throughout design and implementation, reviewing and verifying all AI output (documented in an AI Collaboration Narrative)

Technologies: .NET 10 (C#), ASP.NET Web API, EF Core, PostgreSQL, Docker, xUnit, Testcontainers, Serilog, OpenTelemetry

TECHNICAL SKILLS

- **Back-end:** C#, ASP.NET Core/Web API, Entity Framework, RESTful APIs
- **Database & Messaging:** SQL Server, PostgreSQL, Redis, MongoDB, RabbitMQ
- **DevOps & Tools:** Docker, Git, xUnit, Testcontainers, Swagger, JWT
- **Frontend:** AngularJS, React.js

SOFT SKILLS

- Teamwork
- Problem-solving
- Adaptability
- Time management
- Presentation skills
- Work under pressure
- Responsibility & accountability

LANGUAGE

- **English level:** IELTS (5.5) - capable of everyday communication, reading technical materials, and following conversations.